

Task 7

DDoS Attack Analysis — Five Recent Incidents

1) Cloudflare — Record 22.2 Tbps DDoS (Sept 2025)

Target: Internet infrastructure and Cloudflare customers (hyper-volumetric hit measured at 22.2 Tbps and 10.6 Bpps; short 40-second duration).

Technology: Hyper-volumetric amplification and reflection techniques combined with massive botnet traffic; very high packet-per-second rates.

Motive: Unknown/attribution unconfirmed; likely opportunistic disruptor or competitive/geopolitical actor testing capability.

Impact: Global telemetry spikes, short-duration disruption risk for unprotected services; showcased limits of capacity for many vendors.

Mitigation notes: Massive scrubbing capacity, anycast distribution, ingress filtering, and multi-layer (network + application) defenses were essential.

2) Aisuru botnet multi-Tbps attacks (2024–2025)

Target: Multiple US ISPs, news sites, and gaming services — several multi-Tbps attacks reported (6 Tbps, 11 Tbps, and peaks reported up to ~22 Tbps in later 2025 speculation).

Technology: New large IoT/proxy-based botnet variants leveraging compromised edge devices and reflected amplification; modular and automated attack orchestration. Motive: Likely criminal-to-rent DDoS-for-hire, or disruption for extortion/competitive interference.

Impact: Service disruptions, increased mitigation costs for ISPs and cloud providers.

Mitigation notes: ISP cooperation, upstream scrubbing, botnet takedown efforts, and device patching are key.

3) Gcore mitigates 6 Tbps attack (Oct 14, 2025)

Target: Gcore customers / global infrastructure — a 6 Tbps, 5.3 Bpps event mitigated by Gcore's network.

Technology: Multi-vector volumetric flood (high bandwidth and packet rate).

Motive: Unknown; possibly capacity-testing or criminal campaigns.

Impact: No reported large-scale outages thanks to mitigation; underscores need for large scrubbing capacity.

Mitigation notes: Edge scrubbing, rapid traffic routing to scrubbing centers, and scalable network capacity.

4) Gaming platform disruptions — Steam/Riot (Oct 2025)

Target: Gaming platforms and services including Steam and Riot — service disruptions and outages reported, DDoS suspected.

Technology: High-volume DDoS targeting game APIs, authentication endpoints, or matchmaking servers; likely botnet-driven.

Motive: Competitive disruption, revenge/retaliation (e.g., over game bans), or extortion aimed at gaming companies.

Impact: Player access issues, reputational damage, and increased customer support load.

Mitigation notes: Application-layer protections (rate-limiting, WAF), backend redundancy, and traffic scrubbing.

5) Operation Sindoor — Attacks on India's National Stock Exchange (Oct 2025 report)

Target: National Stock Exchange (NSE) of India experienced extremely high volumes of attacks (reported as 400 million attack events in a single day during 'Operation Sindoor').

Technology: Large-scale scanning, reconnaissance and volumetric attack patterns; distributed probing and flooding.

Motive: Unknown/likely politically motivated or financially targeted reconnaissance and disruption.

Impact: Elevated risk to financial market infrastructure; prompted intensified monitoring by the NSE security teams.

Mitigation notes: Financial organizations need dedicated threat-hunting teams, DDoS scrubbing, and close coordination with ISPs and national CERTs.

Deep-dive: Cloudflare 22.2 Tbps DDoS (selected incident)

Overview:

In early September 2025 Cloudflare (and other industry telemetry) recorded one of the largest DDoS attacks observed to date, peaking at 22.2 terabits per second (Tbps) and 10.6 billion packets per second (Bpps). The event was short (approximately 40 seconds) but represents a new scale in packet and bandwidth floods and demonstrates how modern botnets and reflection/amplification techniques can produce transient but extremely powerful spikes.

1) Target

The attack primarily targeted internet infrastructure and services protected by major content delivery and edge-security providers. While Cloudflare reported the telemetry values, the direct intended victim(s) were not publicly and specifically named in initial disclosures.

2) Technology used (observed/suspected)

- Massive botnet-driven volumetric traffic, combining reflected amplification and direct bot traffic.

- Extremely high packet-per-second (PPS) flood signatures, suggesting use of resourceconstrained devices (IoT, home routers, unmanaged devices) that can generate large numbers of small packets.
- Likely multi-vector orchestration that mixes UDP amplification, spoofed-source floods, and possibly application-layer probes to stress both bandwidth and connectionhandling capacity.

3) Attacker motive

Attribution remained unconfirmed in public reporting. Typical motives that match this pattern include:

- Opportunistic testing of attacker capability or 'flexing' by botnet operators.
- Financial motivation (renting DDoS-for-hire services or extortion) where attackers demonstrate power to coerce targets.
- Geopolitical or protest-driven disruption, though for this specific Cloudflarerecord event, no definitive political claim was publicized.

4) Overall impact

- Short-term spikes in global traffic telemetry and stress on networks and scrubbing capacity.
- Potential service disruption for unprotected or under-provisioned victims during peak seconds.
- Economic and operational cost increase for mitigation partners and downstream providers.
- A strategic signal to the security community that scrubbing capacity and packet-rate defenses must scale beyond previous expectations.

5) Defensive strategies that could have mitigated or reduced impact

Recommended multi-layer defenses (network + transport + application) include: -

Anycasted, distributed scrubbing networks with elastic capacity (use multiple cloud/CDN providers if possible).

- Upstream ISP cooperation and BGP-based traffic rerouting to scrubbing centers.
- Ingress filtering (BCP38) and egress filtering across networks to reduce spoofed traffic.
- Rate-limiting and connection throttling for application endpoints; SYN cookies and robust state handling on edge devices.
- Customer segregation (isolate critical services on dedicated infrastructure) and use of zero-trust access for admin interfaces.
- Regular device patching and reducing Internet exposure of unmanaged IoT devices to limit botnet recruitment.
- DDoS-ready incident playbooks, runbooks, and regular tabletop exercises with ISPs, scrubbing providers, and national CERTs.

Sources and further reading

- Cloudflare / industry reports on 2025 DDoS records (Cloudflare Q2 and Cloudflare telemetry reporting of 22.2 Tbps).
- World Economic Forum and news summaries reporting Cloudflare's 22.2 Tbps observation.
- KrebsOnSecurity reporting on Aisuru and previous record attacks.
- Gcore press release about a 6 Tbps attack mitigation (Oct 14, 2025).
- Tech reports on RondoDox botnet (Trend Micro/TechRadar).
- CyberNews reporting on gaming platform disruptions (Steam/Riot) Oct 2025.
- Times of India coverage of 'Operation Sindoor' attacks on India's NSE.

Summary

This document summarizes five recent, high-profile distributed denial-of-service (DDoS) incidents from 2024–2025 and provides a detailed investigation of one chosen incident: the record-breaking 22.2 Tbps DDoS event reported by Cloudflare in 2025. Each incident summary covers the target, technology used (observed or suspected), attacker motive (as reported or inferred), overall impact, and defensive strategies that could mitigate similar future attacks.